Lab Exercise 18- Volume in Kubernetes Cluster

Volumes in Kubernetes provide a way to store and share data among containers in a Pod. This lab exercise will guide you through the creation and use of different types of volumes in Kubernetes.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

Step 2: Create a Pod with an emptyDir Volume

Create a file named **emptydir-pod.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: test-vol1
spec:
  containers:
    - image: nginx
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      emptyDir: {}
```

In this manifest:

The emptyDir volume is a temporary storage that is created when the Pod is assigned to a node and exists as long as the Pod is running.

Apply the manifest to create the Pod:

```
kubectl apply -f emptydir-pod.yaml
```

Step 3: Verify the Pod and Volume

Check the status of the Pod:

```
kubectl get pods
```

Access the Pod's shell:

```
kubectl exec -it test-vol1 -- bash
```

Inside the Pod, create a file in the /data directory:

```
touch f1 f2 f3
```

You should see the text "Hello, Kubernetes!".

Exit the Pod shell:

```
exit
```

Step 4: Create a Pod with a hostPath Volume

Create a file named **hostpath-pod.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: test-vol2
spec:
  containers:
    - image: coolgourav147/nginx-custom
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      hostPath:
        path: /test
```

In this manifest:

The hostPath volume mounts a directory from the host into the Pod at /mnt/data.
The busybox container writes the current date to a log file every 5 seconds.

Apply the manifest to create the Pod:

```
kubectl apply -f hostpath-pod.yaml
```

Step 3: Verify the Pod and Volume

Check the status of the Pod:

```
kubectl get pods
```

Access the Pod's logs to see the data being written:

```
kubectl exec -it test-vol2 -- bash
```

You should see the dates being appended to the log file.

Check the host directory to verify the volume:

If you have access to the node running the Pod, you can verify the data in /tmp/data/log.txt.

```
C:\Users\NK>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Updating the running docker "minikube" container ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

C:\Users\NK>(echo apiVersion: v1 & echo kind: Pod & echo metadata: & echo   name: test-vol1 & echo spec: & echo   containers: & echo     - image: nginx & ec
ho       name: test-container & echo       volumeMounts: & echo         - mountPath: /data & echo           name: first-volume & echo   volumes: & echo
- name: first-volume & echo       emptyDir: {}) > emptydir-pod.yaml

C:\Users\NK>kubectl apply -f emptydir-pod.yaml
pod/test-vol1 created

C:\Users\NK>kubectl get pods
NAME             READY   STATUS             RESTARTS      AGE
my-pod           1/1     Running            2 (92s ago)   4h8m
pod-with-secret  0/1     CrashLoopBackOff   0             4m57s
test-vol1        1/1     Running            0             8s

C:\Users\NK>kubectl exec -it test-vol1 -- bash
root@test-vol1:/# cd /data
root@test-vol1:/data# touch f1 f2 f3
root@test-vol1:/data# ls /data
```

Command Prompt

```
    Host Port:      <none>
    State:          Running
      Started:      Tue, 28 Apr 2026 12:55:17 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /data from first-volume (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-f7clh (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  first-volume:
    Type:       EmptyDir (a temporary directory that shares a pod's lifetime)
    Medium:
    SizeLimit:  <unset>
  kube-api-access-f7clh:
    Type:                     Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:   3607
    ConfigMapName:            kube-root-ca.crt
    Optional:                 false
    DownwardAPI:              true
QoS Class:                    BestEffort
Node-Selectors:               <none>
Tolerations:                  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  82s   default-scheduler  Successfully assigned default/test-vol1 to minikube
  Normal  Pulling    81s   kubelet            Pulling image "nginx"
  Normal  Pulled     76s   kubelet            Successfully pulled image "nginx" in 4.801s (4.801s including waiting). Image size: 160939956 bytes.
  Normal  Created    76s   kubelet            Container created
  Normal  Started    76s   kubelet            Container started

C:\Users\NK>kubectl describe pod test-vol2
```

```
  IP:  10.244.0.70
Containers:
  test-container:
    Container ID:   docker://347e710491bde2c23aba5ce1d4f3e2d667a90f610917722b2535fc5a3b6de0c1
    Image:          coolgourav147/nginx-custom
    Image ID:       docker-pullable://coolgourav147/nginx-custom@sha256:83c0272ec27d28017925099cdc5c3931c03c2d8b1cf20fcc7ba2f1d01d60bcb4
    Port:           <none>
    Host Port:      <none>
    State:          Running
      Started:      Tue, 28 Apr 2026 12:56:13 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /data from first-volume (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-m7mf8 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  first-volume:
    Type:          HostPath (bare host directory volume)
    Path:          /test
    HostPathType:
  kube-api-access-m7mf8:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
```

```
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  45s   default-scheduler  Successfully assigned default/test-vol2 to minikube
  Normal  Pulling    45s   kubelet            Pulling image "coolgourav147/nginx-custom"
  Normal  Pulled     23s   kubelet            Successfully pulled image "coolgourav147/nginx-custom" in 21.567s (21.567s including waiting). Image size: 187
968468 bytes.
  Normal  Created    23s   kubelet            Container created
  Normal  Started    23s   kubelet            Container started

C:\Users\NK>kubectl delete pod test-vol1
pod "test-vol1" deleted from default namespace

C:\Users\NK>kubectl delete pod test-vol2
pod "test-vol2" deleted from default namespace

C:\Users\NK>kubectl get pods
NAME             READY   STATUS                      RESTARTS        AGE
my-pod           1/1     Running                     2 (3m12s ago)   4h10m
pod-with-secret  0/1     CreateContainerConfigError  0               6m37s

C:\Users\NK>
```